Recall, not verification, is the bottleneck when frontier LLMs elucidate molecular structures from real spectra

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Given the molecular formula together with the infrared band list and ^{1}H/^{13}C shift lists exactly as reported in an open-access paper, how often does a frontier large language model (here, Claude) recover the correct molecular *constitution*? We find **28%** (top-1, n=194; 95% CI 22–35) — far below the near-100% implied by curated demonstrations, and **15%** once the benchmark's deliberate 50/50 difficulty balance is reweighted to the composition of the corpus it was drawn from (§3).

The bottleneck is not the model's judgment but its *proposal*. The model proposes the true structure for only **34%** of compounds; where it does, forward-verification — predicting each candidate's ^{13}C spectrum and re-ranking by agreement with the observed one — selects it **89%** of the time (58/65). **Recall, not verification, is the wall.**

We release three things: **IRexp**, the largest openly redistributable collection of *experimental* infrared band lists (121,233 records, a third structure-linked); **IRSpectra-Bench**, a blind, mechanically scored benchmark of 194 compounds; and a training-free **forward-verification** recipe. The two levers are distinct. Verification re-ranks a fixed candidate set and lifts top-1 from 28% to 30%, while *generating wider* moves recall — 32% to 42%, carrying top-1 from 23% to 30% (§5.2, §5.3); both of those top-1 steps are directional rather than statistically resolved, and the recall gain is the better-supported effect. Fine-tuning a small generator on IRexp moves the wall rather than removing it (§5.6). Masking the spectra drops top-1 from 23% to 5% on the same compounds (§4.6), and Grok 4.6, Gemini 3.7 Flash and GPT-5.6 Sol all verify better than they generate (§4.7). All data, predictions and code are released.

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1. Introduction

Determining a molecule's structure from its spectra is a central, time-consuming task in synthetic and analytical chemistry. Machine learning attacks it with encoder–decoder models trained on paired corpora — *Spectro* learns ^{1}H/^{13}C/IR → SELFIES from 6,833 molecules[1] — and general-purpose LLMs are now reported to do it off-the-shelf: a non-peer-reviewed 2026 industrial white paper found that Claude Opus matched or beat commercial NMR-prediction software in the forward direction (structure→spectrum, $\pm$0.08 ppm ^{1}H) and "recovered all eight simpler structures on every attempt" in the inverse.[2] We treat those numbers as a motivating claim to test against peer-reviewed benchmarks, not as an established baseline.

That evaluation is narrow: 15 inverse problems on curated single-ring or two-fragment molecules, NMR only, seven of them given the *starting-material structure* as a hint, and "recovery" scored leniently over three runs and three ranked candidates. The question a chemist actually faces — *take an arbitrary experimental spectrum from a paper and recover the structure* — is left open. It needs a large, diverse, real benchmark; blind, reproducible scoring; and honest accounting for the methodological choices that inflate or deflate the apparent score.

We provide all three. **IRexp** (§2) is, to our knowledge, the largest openly *redistributable* collection of experimental IR **band lists** (121,233 records; 43,060 structure-linked; 33,201 IR+^{1}H+^{13}C+structure); the superlative is scoped to that object type, since view-only libraries such as SDBS and commercial libraries hold more structure-linked *spectra* but cannot be redistributed (§2.1). On it, **IRSpectra-Bench** (§3) is, to our knowledge, the first blind, mechanically scored, complexity-stratified evaluation of frontier-LLM elucidation on real spectra: measured in depth on Claude, replicated across three other model families (§4.7), and isolating a large solver-methodology effect with a within-compound control (§4). Its ground truth is literature structures resolved deterministically (OPSIN/RDKit) and checked mechanically against the source articles (560/560 bands on a seed-fixed sample, §2.3; the *expert-chemist* review is prepared but not yet run, §7); no LLM curates labels or scores predictions.

Forward-verification elucidation (§5) turns the model's *strong* direction (forward prediction) against its *weak* one (inverse regiochemistry); the loop is prior art (§1.1), the decomposition it enables is ours, run training-free over **every** benchmark compound. The finding is a sharp asymmetry: given a candidate set containing it, the model *verifies* the correct structure 89% of the time, yet *proposes* it for only 34% of compounds — recall, not verification, is the wall (Fig. 1). The gain is bounded (top-1 28%→30% on n=194; 23%→30% on the 60-compound arm where wide generation was also tested) and, by our own measurements, cannot exceed a recall/precision ceiling without sharper verification or 2D-NMR data.

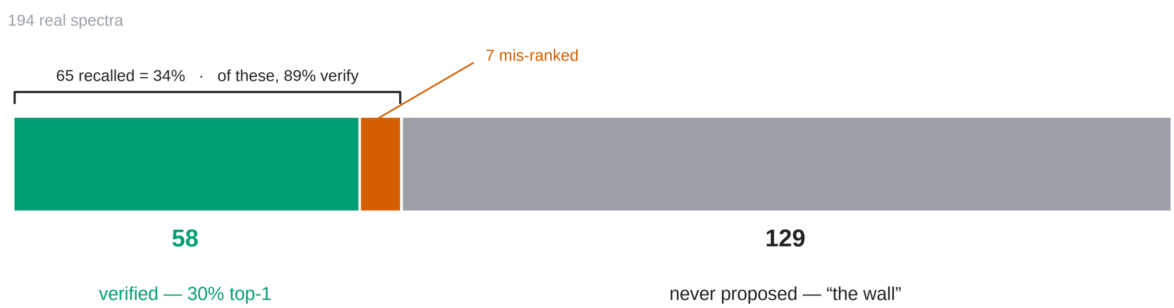

Fig. 1: All 194 benchmark compounds as a single part-to-whole bar. The true structure is verified top-1 for 58 (green), recalled but mis-ranked for 7 (vermilion), and never proposed for 129 (grey) — *the wall*, 66% of the bar. Generation recall is 65/194 = 34%; conditional verification precision is 58/65 = 89%; their product is the 30% end-to-end top-1. The two rates have different denominators and are not differenced.

Solver and verifier (Fig. S1) are LLM agents on a consumer subscription — no fine-tuning, no API spend — cheap to *re-run* but not reproducible: the harness pins no model snapshot, exposes no temperature or seed, and carries an undisclosed product system prompt, so an outside group reproduces it distributionally, not exactly (§8). Scoring is exactly reproducible: predictions, ground truth and scorers are released, and every number in the training-free core (§3–§5.3, §4.6) regenerates from them, except §5.4's HOSE lookup and GNN, which need an nmrshiftdb2 dump we cannot redistribute (see *Data and code availability*); that GNN and the §5.6 generator are trained probes, fenced as complements to the training-free core.

1.1 Related work

Trained spectra→structure models. The dominant line trains sequence or graph decoders on spectra: *Spectro*[1], a multitask CNN+transformer for routine 1D-NMR[3], set/graph transformers such as NMRTrans[4], and — closest to our multimodal setting — **NMIRacle**[5], conditioned jointly on IR + ^{1}H + ^{13}C. They are accurate *in-distribution* but retrained per modality and dependent on a labelled spectra→structure corpus. That resource is scarce rather than absent, and one concurrent effort attacks it the same way we do: NMRTrans builds NMRSpec, a literature-mined corpus of experimental ^{1}H/^{13}C spectra[4]. IRexp is the complement rather than the competitor — infrared band lists, released under permissive licences for redistribution (§2.3) — and the two together are the open, experimental, multimodal corpus neither supplies alone.

LLMs as elucidators, and how this work differs. LLM-orchestrated tools already plan and execute real syntheses (ChemCrow[6], Coscientist[7]); such systems *call* an elucidation routine, and this paper measures what one returns. General-purpose LLMs have been applied off-the-shelf[2], as have multimodal models over multi-spectral input (SpectraLLM, SpecMol)[8,9] and knowledge-enhanced tree-search reasoning[10]; **MolPuzzle**[11] supplies IR+MS+^{1}H+^{13}C puzzles with the formula given, and **IR-Agent**[12] emulates expert IR interpretation on experimental spectra with a multi-agent framework. We claim priority on none of them: IR-Agent improves *how the model reads a spectrum*, where we ask what limits the outcome once it has.

Prior benchmarks and trained baselines evaluate on simulated or software-predicted spectra[1,5] or on curated single-instrument libraries — Alberts et al. use NIST gas-phase IR[13,14] — where IRSpectra-Bench uses spectra **as reported by the authors of the source papers**, across thousands of laboratories and instruments; the contrast is with both simulation *and* curated uniformity. That line is also the state of the art for *trained, formula-conditioned IR* elucidation — an IR→structure transformer[13] now at **63.8% top-1 / 84.0% top-10 on experimental NIST gas-phase spectra** given the formula, pretrained on 1,399,806 simulated spectra[14]. That headline is on a **6–13-heavy-atom** subset, the range where our own accuracy is highest (60.5% top-1 for ≤15 heavy atoms, §4.1), so on comparable molecules the training-free LLM and the purpose-trained transformer are closer than the headline numbers suggest. Size is not the whole story, though, and we do not claim it is: the same work reports **59.94%** on a 5–35-heavy-atom set (n=5,024), which spans most of our range and falls only four points. What differs there is the data, not the size — a single-instrument gas-phase library against heterogeneous literature-reported band lists. Neither setting bounds the other. Reported accuracies also swing with inference method and scoring harness: GPT-4o is scored at **1.4%** on MolPuzzle by the benchmark's own authors[11], at **27.8%** under a plain chain-of-thought harness, and at **57.8%** with knowledge-enhanced tree-search reasoning[10] — a factor of forty for one model on one benchmark. Cross-paper numbers are therefore not comparable, and we fix and release a single pre-registered RDKit-InChIKey protocol with bootstrap CIs.

Computational NMR for structure validation. Forward verification is the LLM analog of a workflow chemists already trust — compute the spectrum each candidate *would* give and match it to experiment: DP4 / DP4+ over GIAO-DFT shifts in solution[15,16], **NMR crystallography** with GIPAW shifts in the solid state[17,18]. We swap the quantum-chemical predictor for a forward LLM, trading accuracy for zero setup cost, and inherit the principle that *verification by forward prediction is easier than inverse generation.* **Computer-assisted structure elucidation** (CASE) has run the loop since the 1990s and overturned published assignments[19,20]; its generator is a strict enumerator with exhaustive recall by construction[20] — the axis on which we find the LLM weak, and the axis existing LLM benchmarks leave unmeasured by reporting a single aggregate score. The decomposition is stable under every perturbation we could run — four Claude models, a second chemical domain, four verifiers, all recall-bound (§4.4, §4.5, §5.4) — but is not tested outside the Claude family (§7).

The generate-and-forward-verify loop is not ours. NMR-Solver[21] builds it without an LLM, retrieving and fragment-recombining candidates and ranking them by ^{1}H/^{13}C shifts forward-predicted with NMRNet (^{13}C MAE 1.098 ppm); on ~450 experimental literature spectra with the formula supplied it reports **52.89% top-1**, well above our 28.4%. §5 asks a different question — *how much of the remaining error is generation and how much is verification* — answered by decomposition (§5.2) rather than by a better solver. That its predictor is roughly twice as sharp as the ~2 ppm LLM forward-predictor §5.1 identifies as the binding constraint, and delivers roughly twice the top-1, is what our §5.4 ablation predicts and §5.1's separability measurement implies. **NMRAgent**[22] couples spectral tools to a knowledge graph and validates on newly isolated natural products; it is complementary — a best-effort system, where we measure an off-the-shelf model.

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2. The IRexp dataset

2.1 Motivation

An IRexp record is a **band list** — peak positions in cm^{-1} transcribed by an author into a paper's text — with co-reported ^{1}H/^{13}C shift lists and, where resolvable, the 2D structure. **IRexp contains no absorbance traces.** That is the published form, and what a language model consumes, but it is a different object from a digitised spectrum and the two should not be counted against one another.

Among *digitised spectra*, free downloads run to the NIST WebBook[23] (~16k) and the Chemotion ELN deposit[24] (~2k); AIST SDBS[25] is larger (~54k FT-IR, all structure-linked) but *view-only* (50 spectra/day, no bulk export), so **IRexp does not compete there: SDBS alone holds more structure-linked spectra than IRexp holds structure-linked band lists.** Among text-derived *band lists* IRexp is the largest *openly redistributable* collection by record count, a claim deliberately scoped to that object type.

2.2 Construction

Experimental sections report per-compound IR band lists with co-reported ^{1}H/^{13}C NMR in a stable textual convention. Mining it is mature[26]; we add a pipeline specialised to the *IR band list*, the modality those tools passed over.

- **Discovery.** Open-access literature is harvested in bulk from the PMC Open-Access Subset[27] and the Chemotion FT-IR deposit (RADAR4Chem).
- **Extraction.** A deterministic parser segments experimental text per compound and extracts IR wavenumbers and ^{1}H/^{13}C shifts, gated against scan-range artefacts and prose false-positives.
- **Resolution.** IUPAC names go to SMILES with OPSIN[28] and RDKit[29] canonicalisation (InChIKey, SELFIES[30]), with a PubChem[31] fallback for trivial names; handling open-access label conventions and narrative artefacts raised structure coverage from 24% to **35%**.

2.3 Contents and licensing

Table 1 gives what the release holds and where each part came from.

Table 1. IRexp dataset contents and provenance.

field	value
experimental IR band-list records	**121,233**
…co-reporting ^{1}H and/or ^{13}C NMR	87,075 (72%)
…with a resolved 2D structure	**43,060 (35.5%)**
…of these, with ^{1}H and/or ^{13}C NMR	**40,702**
…of these, with both ^{1}H and ^{13}C (full quadruples)	**33,201**
by provenance: author-transcribed from PMC-OA text (CC-BY)	119,345
by provenance: peak-picked from Chemotion ELN deposits (CC-BY-SA)	1,888

The pools differ: PMC records are author-transcribed (median 9 bands), Chemotion records peak-picked from deposited spectra (median 39), so users training on band density should treat them apart. `scripts/split_license_pools.py` separates them on `source_doi` and stamps each record's licence.

Extraction fidelity is measured, not assumed. On a seed-fixed random sample of 60 PMC-sourced records (`scripts/audit_extraction.py --n 60 --seed 0`) we re-fetched each article and checked every recorded wavenumber against its text: **560/560 bands and 60/60 records were confirmed** (Wilson 95% CI 99.3–100% and 94–100%). This bounds *transcription* error — hallucinated, mis-parsed or unit-mangled values — below 1% of bands. It does **not** measure whether the parser found every IR string in every paper, a recall question requiring human reading; that audit is prepared but not yet run (§7).

`irexp_resolved` (43,060 records, 100% structure-linked) is the benchmark-ready split, ~6× the 6,833-molecule set used to train Spectro[1] (Fig. S2).

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3. Benchmark design (IRSpectra-Bench)

From `irexp_resolved` we draw **IRSpectra-Bench**, 194 blind elucidation problems, each giving the **molecular formula** (as from HRMS), the **IR band list** and the **^{1}H and ^{13}C shift lists**; no name, SMILES or hint. Main-round ground truths are **spectrally validated** by an automated RD-Kit check (^{13}C peaks vs symmetry-unique carbons, formula match, SELFIES round-trip) excluding merged or incomplete spectra — 6/140, leaving 134. `scripts/validate_benchmark.py` runs the deterministic filter over all three rounds, regenerating every `clean_qids.json` from the released questions and answers.

The filter does not gate on ^{1}H, and we report what that leaves. In **13 of the 194** retained records the reported ^{1}H integral exceeds the reference structure's hydrogen count — residual solvent, water, exchangeable protons, or a rotamer mixture. They are printed as a diagnostic, not excluded: the cohort was fixed in advance, and re-filtering post hoc on a criterion chosen after seeing results is a degree of freedom a benchmark should not take. We report the sensitivity instead: dropping all 13 moves the headline from **28.4% to 29.3%** (53/181), +0.9 points, leaving every conclusion unchanged. The controlled rounds' 60 compounds are fixed, pre-registered sets, audited the same way but reported rather than filtered (57/60 pass; §7).

Difficulty is a declared property of the compound, not a post-hoc label. By RDKit ring analysis a compound is **simple** iff it has at most two rings, no fused/spiro/bridgehead system and $\leq$22 heavy atoms; all others are **complex** (98 simple / 96 complex), so the split is exhaustive. The threshold is not load-bearing: sweeping it from 18 to 26 moves the simple-minus-complex top-1 gap only between 36 and 40 points (39.6 as released; `scripts/difficulty_sensitivity.py`). This axis is distinct from the continuous size gradient of §4.1 ($\leq$15 / 16–25 / >25 heavy atoms), and InChIKey de-duplication across rounds prevents leakage.

The 50/50 balance is by design, and the corpus is not balanced. The sampler fills each stratum to half the round, so "98 simple / 96 complex" is a property of the draw and not an observation about the literature. Under the same eligibility filters the eligible corpus is **17.5% simple and 82.5% complex** (5,059 / 23,929 of 28,988; median 26 heavy atoms against the benchmark's 20), so simple compounds are enriched **2.9$\times$** (`scripts/corpus_reweight.py`). Balancing is the right choice for measuring a gradient — an unbalanced draw would put almost no mass in the stratum where the model succeeds — but it means the headline is a weighted average under weights we chose. Reweighted to the corpus, top-1 falls from 28.4% to **15.2%** (95% CI 10.7–20.4) and recall from 33.5% to **19.8%** (14.3–25.7). We report the benchmark figure as the headline because it is the one the released set reproduces, and the reweighted figure as the better estimate of what a chemist meets on an arbitrary paper. The reweighting assumes the one sampler filter it cannot apply — recovery of the raw ^{1}H string, which needs a network fetch per record — is independent of difficulty.

Scoring is mechanical. A prediction is *correct* if its RDKit InChIKey connectivity layer (first 14 characters) matches the reference: we score *constitution*, so correct constitution with wrong stereochemistry counts as correct. We report the strict alternative rather than assert it is immaterial — the *full*, stereochemistry-sensitive InChIKey gives **21.1% top-1 and 25.8% recovered** (41/194 and 50/194) against 28.4% / 33.5%, 7.3 points lower, so the constitution figure is an upper bound on full-stereochemistry accuracy. Constitution is the headline because 1D ^{1}H/^{13}C/IR rarely fixes absolute configuration: only 10.3% (20/194) of answers carry a *defined* (assigned R/S) stereocentre, and a model is penalised there for information the prompt never contained (`scripts/score_main.py --stereo`).

Metrics. Top-1 (exact constitution) is the fraction whose best-ranked candidate matches at the connectivity layer; **recovered (top-3)** the fraction whose reference appears among the up-to-three returned candidates, the lenient "recovery" protocol of ref.[2] — their metric, under their name. **Generation recall** is the fraction whose reference is in the candidate pool *before* re-ranking, the ceiling any verifier can reach; it coincides with recovered (top-3) except in the generate-wide arm of §5.3, where the pool is larger. **Conditional-on-recall precision**, over recall-positive compounds alone, isolates verification from generation. The forward verifier ranks by a symmetric **chamfer distance** between predicted and observed ^{13}C peak sets (lower is better, no equal-count requirement); Morgan(2, 2048)[32] Tanimoto gives a graded scaffold-family signal. CIs are bootstrap 95%; model-vs-model differences use McNemar's exact test[33] with Holm correction[34].

Solvers are frontier LLM agents (Claude Opus), one sub-agent per batch, closed-book under a consumer subscription: no tools beyond an RDKit formula check, no ground truth, verified by grep-auditing transcripts.

Formula adherence is high but imperfect — 91.3% (115/126) of forward-verification candidates match the given formula exactly, 76.6% across the full top-3 pool — and **not uniform across rounds**: on top-1 answers it is 95.0% (38/40) in v3 and 90.0% (18/20) in the v2-control, against **77.6% (104/134) in the headline main round** — a structure of the wrong composition, in violation of a constraint the solver was explicitly handed, for about one main-round compound in five. Such answers score as misses, so this inflates nothing, but the main-round arm is the weakest-constrained (`scripts/analyze_misses.py`).

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4. How well do LLMs elucidate real structures?

4.1 Headline performance

Solver agents work blind from formula + IR + ^{1}H + ^{13}C and return up to three ranked candidates, in bounded contexts reset every 2–12 compounds rather than one long context — the variable §4.3 shows matters. The benchmark is all **194 compounds** (134 spectrally-validated + 60 controlled-round; `data/benchmark_main/raw/`).

Table 2. Headline elucidation performance on IRSpectra-Bench (n=194), overall and by difficulty stratum, with bootstrap 95% CIs.

metric	overall (n=194)	simple (n=98)	complex (n=96)
top-1 exact constitution	**28.4%** [22–35]	48.0% [39–57]	8.3% [3–15]
recovered (within top-3)	33.5% [27–40]	54.1% [44–63]	12.5% [6–20]
scaffold-level (best Tanimoto $\geq$ 0.45)	56%	73%	39%
mean best Tanimoto	0.59	0.73	0.45

A strictly-validated subset with the controlled rounds restricted to their clean parts too (134 main-clean + 57 controlled-clean = 191) reproduces this within ~1 point on every metric — top-1 **28.8%**, 95% CI 23–36; recall 34.0%; simple 49.0% / complex 8.4% — so the asymmetric inclusion of the pre-registered controlled sets does not drive it. What does move the overall figure is the deliberate 50/50 difficulty balance: reweighted to the 17.5%-simple composition of the eligible corpus, top-1 is **15.2%** [10.7, 20.4] and recall **19.8%** [14.3, 25.7] (§3). The per-stratum figures are unaffected — only the weights change — and the reweighted number, not the headline, is the one to read as performance on an arbitrary paper. Accuracy falls monotonically with size in step with the difficulty gradient (Fig. 2): **60.5%** top-1 at $\leq$15 heavy atoms, 28.3% at 16–25, **7.0%** above 25 (Fig. S3).

Of the 137 analysable top-1 misses (139 in all; two predictions did not parse; `scripts/analyze_misses.py`), **76.6% are constitutional isomers of the true structure** against 23.4% with the wrong formula. That is not mostly regiochemistry: only **22.6%** share the true Murcko scaffold, 2.9% reach Tanimoto $\geq$ 0.85, and the median isomeric-miss Tanimoto is 0.39 — strict regiochemistry is a fifth of failures, not the bulk, most misses being larger rearrangements of the same atoms. Near-degenerate ^{1}H/^{13}C shifts under-determine connectivity, hence HMBC and NOESY; the honest statement is *wrong connectivity at the right composition.*

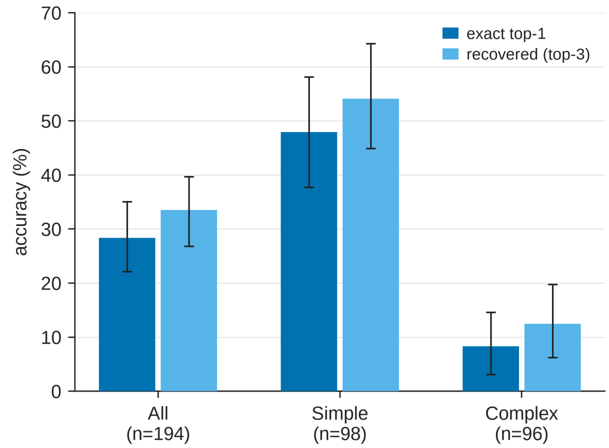

Fig. 2: Top-1 and recovered accuracy on IRSpectra-Bench by difficulty (all / simple / complex, n=194), with bootstrap 95% confidence intervals. The benchmark separates a realistic difficulty range: simple targets are solved four to six times as often as complex ones on both metrics.

4.2 Reconciling with prior reports

Our 28% top-1 sits far below the ~100% on "simple" molecules reported for the same model class in a non-peer-reviewed company white paper.[2] Unscored independently, it is best read against the *peer-reviewed* record: MolPuzzle[11] and its re-scorings[10], and the trained baselines[1,5]. Four methodology and scoring choices differ, each in the direction that raises a reported number; we name them without apportioning the gap. **Difficulty:** ring count bounds difficulty from below rather than proxying for it — our simple stratum includes a hexasubstituted benzene, while $\geq$4 rings mark 38% of recall-negative against 3% of recall-positive compounds (§5.3). **Scoring:** prior work counts a recovery if the reference appears among three ranked candidates over three independent runs; we report single-run top-1 and top-3. **Hints:** prior hard targets got the starting-material SMILES and we give none, though the formula we do give is worth 5% top-1 alone with the spectra masked (§4.6). **Curation:** prior compounds were hand-selected; ours scraped, unfiltered for solvability.

Versus trained models — a bound, not a leaderboard. No system has been scored on our test set, and published numbers differ in the three respects that most move the score: spectrum realism, hints, exact-match definition. Most trained baselines report in-distribution accuracy on simulated spectra; Alberts et al. do not — 63.8% top-1 from IR alone, formula supplied, on **experimental** NIST gas-phase spectra of a curated single-instrument library of **6–13-heavy-atom** molecules[14]. Ours span 8–60 heavy atoms (median 20), only **15 of 194 (8%)** in that window; against our $\leq$15-heavy-atom stratum (60.5%) their 63.8% is a near-tie. Size alone does not explain the rest: the same work reports **59.94%** over 5–35 heavy atoms (n=5,024), a range that covers most of ours. The remaining difference is what the spectra are — one gas-phase instrument against thousands of laboratories — and the comparison bounds the gap between those regimes rather than ranking the two systems. Multimodal systems report more still, on data of their own making: Spectro **93%** (**82%** with fixed embeddings) on a split whose IR is plotted from reference data and whose NMR is software-*predicted*[1]; NMIRacle 48% top-1 / 66% top-15 on molecules held out from a *simulated* corpus inside its own training distribution, as its authors note[5]. Ours is 28.4% top-1 (33.5% top-3), 29.9% with forward verification (§5), on **blind, real, literature-mined experimental** spectra of out-of-distribution compounds.

One asymmetry cuts against us: NMIRacle takes **no molecular formula** where we supply it (§3), so the settings differ on opposite axes — our spectra real and out-of-distribution against their simulated and in-distribution, their input strictly harder — and neither number bounds the other. Read only as a *bound on the simulated-to-real gap*, the contrast suggests high in-distribution accuracies substantially overstate real-world performance. Metric instability reinforces the caution: the ~40× MolPuzzle swing in §1.1 (GPT-4o: 1.4%[11] to 27.8% to 57.8%[10], method- and harness-dependent) bounds the weight any unaudited near-100% claim[2] can bear.

4.3 Methodology dominates: a within-compound control

The same 20 molecules were solved two ways: (a) one LLM context handling all sequentially with no tools, (b) four independent agents of five compounds each with RDKit formula-checking. **Recovered (top-3)** rose from **5% to 15%** (1/20 → 3/20) and top-1 from 0% to 15% (0/20 → 3/20). McNemar's exact test (§3) does not reach significance, and at this size it could not have: the exact test conditions on the discordant pairs, of which the top-1 arm has three (b=0, c=3, **p=0.25**) and the recovered arm four (b=1, c=3, **p=0.625**). Below six discordant pairs the smallest attainable two-sided p is above 0.05, so "not significant" here reports the design, not the data. The 15% carries a Wilson 95% CI of roughly 5–36%. Bounded, frequently-reset contexts with tool access therefore appear to raise measured performance, consistent in direction but **not established in size** at n=20 (p=0.25): a directional within-compound demonstration, not a measured 3× effect. Small rounds swing widely (15–40% across n=20–40 draws), hence the full **194-compound** headline. It plausibly explains *part* of the gap to optimistic prior reports, whose per-problem API calls implicitly used method (b) — a hypothesis, not a quantified contribution.

4.4 Model comparison: the benchmark ranks capability (and separates the extremes)

We solved a fixed 24-compound subset blind with four Claude models spanning a wide capability range, including the newest (Fable 5), under one prompt, one scorer and one candidate budget (Fig. 3, Table 3).

One protocol asymmetry must be disclosed: the three comparison models solved the 24 compounds as four six-compound contexts each, whereas the Opus column reuses the headline run, where those items sat in one six-compound and two twelve-compound contexts — the variable §4.3 measures at 5%→15%. The Opus estimate is therefore not protocol-matched and, if anything, handicapped by the longer contexts; this touches neither the nesting nor the Fable-vs-Haiku contrast, and a clean re-run is outstanding in `docs/MODELS.md`.

Table 3. Four-model comparison on a fixed 24-compound subset under the identical blind protocol.

model	top-1	recovered	top-1 95% CI
Claude Fable 5	**46%**	54%	[25, 67]
Claude Opus	25%	29%	[8, 42]
Claude Sonnet	21%	25%	[8, 38]
Claude Haiku	**0%**	4%	[0, 14]

CIs are bootstrap except Haiku's: Clopper–Pearson exact for 0/24, where the percentile bootstrap is degenerate.

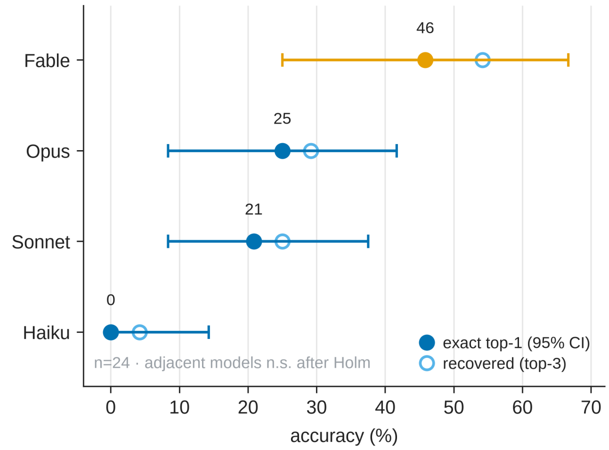

Fig. 3: Four-model comparison on a fixed 24-compound subset, solved blind under one protocol: Fable 5 46% > Opus 25% > Sonnet 21% > Haiku 0% top-1. The outcomes are strictly nested — each stronger model solves a superset of the weaker one's compounds — so the benchmark is capability-sensitive, but at n=24 it is underpowered to separate adjacent models (§4.4).

The models rank in **monotonic capability order** with **strictly nested** outcomes (Haiku ⊂ Sonnet ⊂ Opus ⊂ Fable), so the benchmark is capability-sensitive. The nesting makes the *ranking* robust, but at n=24 the subset is **underpowered to separate adjacent models**: by McNemar's exact test only Fable-vs-Haiku survives multiple-comparison correction (Holm-adjusted p=0.006); Fable-vs-Opus does not (uncorrected p=0.063 — which is the *floor* for its five discordant pairs, so the comparison had no way to reach 0.05 — Holm-adjusted p=0.19), and Opus and Sonnet are indistinguishable. Two mid-tier models agreeing closely puts the recall-bound regime of §4.1 beyond a single model; the newest nearly **doubles** the next-best top-1 yet still misses the majority, leaving the benchmark far from saturated. We used Claude-family models because they are callable for free under one subscription; a cross-vendor sweep needs API access we did without, though the large monotonic spread makes it unlikely the pattern is lineage-specific. ### 4.5 Domain case study: battery-electrolyte chemistry

To test the recall-bound regime on battery-electrolyte chemistry rather than on organic molecules at large, we curated **IRSpectra-Bench-Electrolyte**: a 48-compound subset of IRexp records (eight per class; two yielded no parseable candidate, leaving **46 scored**) across the six functional families that dominate lithium- and sodium-battery electrolytes, held out of every other split and solved blind under the identical protocol. These are literature compounds, **not** operando or in-cell spectra, and we make no claim that they are.

Performance lands in the **same broad regime as the headline benchmark — top-1 26%, recovered 28%** (12/46 and 13/46; at n=46 the 95% CI is wide and overlaps the headline) — consistent with a bottleneck in the elucidation task rather than in any one chemical neighbourhood. The true structure enters the candidate set for only 13 of 46 compounds and ranks first in 12 of those 13. Forward-verification was **not** run here, so this is the recall-bound pattern under the solver's own ranking, not the §5 measurement (Fig. S4). The per-class breakdown is Table 4; at eight compounds a class the intervals overlap throughout, so the ordering is hypothesis-generating rather than an established ranking.

Table 4. Per-class performance on IRSpectra-Bench-Electrolyte (n=46).

electrolyte class	n	top-1	95% CI	recovered (top-3)	95% CI
sp^3-C–F	8	**50%**	[22, 78]	50%	[22, 78]
carbonate	7	29%	[8, 64]	43%	[16, 75]
phosphoryl	8	25%	[7, 59]	25%	[7, 59]
glyme / oligoether	7	29%	[8, 64]	29%	[8, 64]
sulfonyl / sulfonate	8	**12%**	[2, 47]	12%	[2, 47]
nitrile	8	**12%**	[2, 47]	12%	[2, 47]

The intervals overlap almost completely and the spread is within sampling noise (six-class χ^2 p≈0.56; best-vs-worst Fisher exact p≈0.28), so the ordering is hypothesis-generating rather than an established ranking, though chemically legible: C–F couplings pin sp^3-C–F regiochemistry, while sulfonyl and nitrile targets turn on oxidation-state ambiguity and heteroaromatic substitution that ^{1}H/^{13}C shifts underdetermine. It is the recall-bound failure of §4.1 in a domain-specific guise, and it is where §5's **forward-verification** recipe plays a role *analogous* to (not a replacement for) computational-NMR validation.

4.6 Is the model reading the spectra? A formula-only control

Every benchmark compound is mined from open-access literature, so a frontier model may have met it in pretraining — a well-documented hazard for LLM evaluation.[35] We reran the identical blind protocol with every spectral channel masked, leaving the solver the molecular formula and nothing else, which does not determine constitution: accuracy materially above the floor would indicate recall, not reasoning.

The arms were not generated together: the **formula-only** arm was fresh (2026-07-28), the **full-modality** arm archived from June. The bias therefore runs *toward* formula-only — fresh agents reason harder per compound than the archived batched run — and it still collapsed to 5%; a confound that works against the finding cannot manufacture it.

Table 5. Formula-only control, paired on the same 60 compounds as §5.

condition	top-1	recovered (top-3)
formula only	**3/60 (5%)**	3/60 (5%)
formula + IR + ^{1}H + ^{13}C	**14/60 (23%)**	19/60 (32%)

The outcomes are perfectly nested: **eleven** compounds are solved with the spectra and not without, and **none** the other way round (McNemar exact p=0.001). Removing the spectra removes the result.

A second, independent control: publication recency. The formula-only arm shows the spectra carry the signal, not whether memorisation contributes to the part they explain; if recall drove the headline number, accuracy should fall with recency. Across **all 194** compounds, source papers spanning 2008–2026, accuracy is flat: **28.6%** for the older half (≤2020, n=112) against **28.0%** for

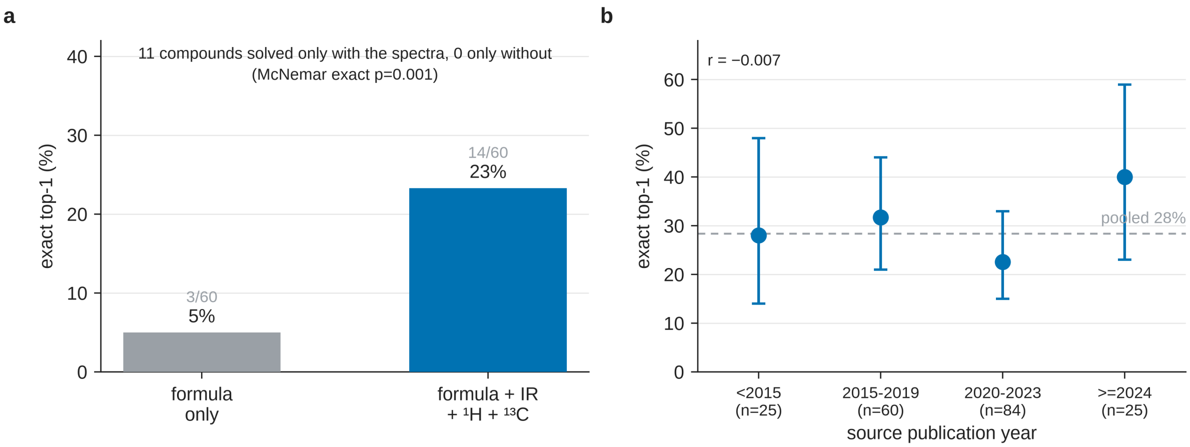

Fig. 4: Two contamination controls. (**a**) The solver reaches 3/60 on formula alone against 14/60 with IR + ^{1}H + ^{13}C, nested outcomes: 11 solved only with the spectra, none only without (McNemar exact p=0.001). (**b**) Accuracy against source-paper year (n=194, Wilson 95% CIs) is flat, point-biserial r = −0.007, where pretraining recall predicts a decline.

the newer (n=82), a point-biserial year–correctness correlation of **r = −0.007**, and no monotone trend across buckets (Fig. 4b), the most recent (≥2024, n=25) being highest at 40% [23, 59].

Newer papers skew larger (median 22 heavy atoms against 20) and size dominates accuracy, biasing the raw split *against* them (§4.1); size-adjusted, the older-minus-newer difference is **−5.1 points, 95% CI [−17.2, +7.0]** (Cochran–Mantel–Haenszel χ^2=0.42, p=0.51, continuity-corrected). Its sign is opposite to what pretraining recall would produce, but the interval comfortably includes zero, so this **bounds** any recency effect rather than demonstrating a reversed one, and by the standard §5.3 applies to its own adjacent conditions we do not read it as directional.

The 5% is not zero, and its three compounds do not all read the same way: one is **absent from PubChem** and near-determined by its benzyne-precursor composition, while a catalogued sulfonamide and a named natural product are as consistent with recognition as with inference. Either way, formula-level recall accounts for at most about a fifth of the headline accuracy on this set. The two controls are independent and agree, but neither is randomised — publication year is observational, and the formula-only arm cannot distinguish memorisation from inference on a near-determining formula — so we claim a strong bound rather than exclusion (§7). Nor can either control address retrieval on the spectral string itself, which the prompt carries verbatim from the source article; §7 states that objection and names the control that would settle it.

4.7 Does the diagnosis hold outside one vendor? A four-vendor replication

Every number so far comes from one lineage, the external-validity gap §7 (iii) named as most important. Six non-Claude models solved the same 60 compounds as Table 7's first column under the identical blind protocol; the three whose output was well-formed enough were carried through forward verification (Table 6).

Table 6. Cross-vendor decomposition on the 60-compound arm. Recall and precision have different denominators, so the criterion is the inequality, not a difference.

model	generation recall	verification precision | recall	multi-candidate only
Claude Opus (Table 7)	19/60 = 32% [21, 44]	16/19 = **84%** [62, 94]	10/13 = 77%
Grok 4.6	32/60 = **53%** [41, 65]	20/32 = 62% [45, 77]	20/32 = 62%
Gemini 3.7 Flash	30/60 = **50%** [38, 62]	22/30 = 73% [56, 86]	22/30 = 73%
GPT-5.6 Sol	25/60 = **42%** [30, 54]	17/25 = 68% [48, 83]	16/24 = 67%

The inequality holds in every arm (Fig. 5): verification precision exceeds generation recall for four independent model families. Recall and precision are measured on the same compounds, so what carries the claim is the paired difference, not whether two marginal intervals overlap. Bootstrapping compounds (`scripts/cross_vendor_gap.py`) resolves three of the four: Claude, Gemini **+23.3 points [+2.9, +44.3]** and GPT-5.6 Sol **+26.3 [+4.2, +49.0]** all exclude zero. Grok does not — **+9.2 [−10.7, +30.3]** — and we report its gap as directional.

Two findings sit underneath. Six of Claude's nineteen recall-positive compounds carried one candidate, so nothing was ranked and any verifier scores them by construction — as §5.2 says, but only against itself — whereas the newer models almost always return three (0–1 singletons), and where a choice existed the four precisions fall in a band that dissolves the apparent Claude advantage. And verification looks vendor-independent where generation does not, precision on multi-candidate sets spanning 62–77% while recall spans 32–53%: the wall is the generator. Three models beat ours on generation recall, which is the diagnosis working: recall is the movable factor.

The candidate budget is not matched, and the correction runs both ways. Recall above asks whether the true structure is anywhere in the candidate list, and the lists differ in length: ours holds **2.20** candidates per compound on this arm against exactly **3.00** for every comparison model (Composer 2.82). A longer list can only raise recall, so the comparison is tilted toward the other vendors — the mirror of the singleton effect we correct on the precision side. At **one** candidate, the only budget every model met, recall is 14/60 = 23% for ours against 23/60 = 38% for Grok, 23/60 = 38% for Gemini and 21/60 = 35% for GPT-5.6 Sol (`scripts/cross_vendor_budget.py`). The ordering survives and three models still lead ours, but Grok's margin is **15 points, not 21**, and 21 is the figure a budget-matched comparison does not support.

Three weaker models are excluded from the decomposition: Composer 2.5 (20% recall) and GPT-5.6 Luna (15%) match the given formula only 67% and 76% of the time, and `nvidia/nemotron-3.5-lightning` managed 2%. §4.1's scorer now prints formula adherence beside recall. A seventh arm, DeepSeek V4 Pro, returned answers for only **18 of 60** compounds before the run ended; the true structure appears for 8 of those, so its recall is a lower bound and it is excluded from the comparison rather than from the release — both its solve and verify files are deposited.

Contamination was controlled, not assumed. Blindness rested on an instruction: these runs used cloud agents with repository access to tracked answer keys. Grok re-solved all ten batches from a clone with the answer files removed: recall 28/60 against 32/60, paired McNemar **p=0.39**, formula adherence 97% against 95% of candidates (`scripts/cross_vendor_sweep.py`). The asymmetry is decisive — a key-reader would solve a *superset*, yet **four compounds were solved only in the arm that had no key**, with 24 of 60 solved by both, which is sampling noise, not copying. And on constitutionally correct structures whose reference carries assigned stereocentres, which 1D spectra cannot supply, Grok reproduced 0/3 correct descriptors, Gemini 0/2 and GPT-5.6 Sol 0/2.

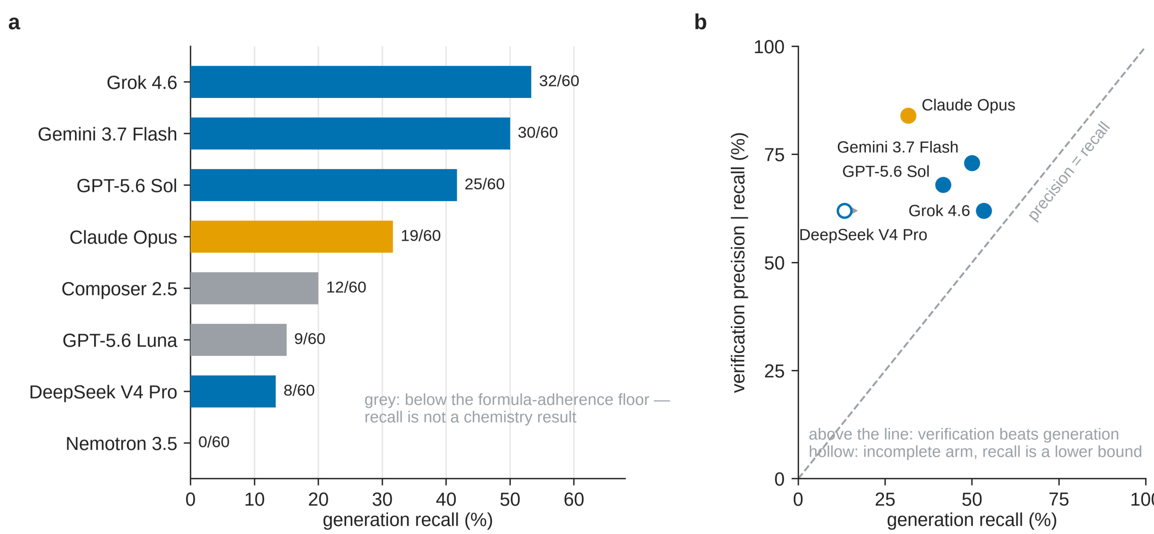

Fig. 5: Every model measured on the 60-compound arm. (**a**) Generation recall, each model at the candidate budget it used — 3.00 candidates per compound for all but ours (2.20) and Composer (2.82), so this axis favours the models that always returned three; grey marks models below the formula-adherence floor, whose recall is not a chemistry result. (**b**) The claim itself: verification precision against generation recall, with the diagonal. Every model sits above the line, so verification is better than generation for all four families; hollow marks an incomplete arm, where recall is a lower bound. The four Claude models of §4.4 are deliberately absent — they ran a different 24-compound subset, and putting them here would imply a comparison that was never made.

Two limits. The models were reached through a coding-assistant harness exposing reasoning-effort tiers, a decoding control the Claude arm never had (§8), so these measure a model *as served*, not a bare endpoint. And the clean-clone control covers only Grok; Gemini and GPT-5.6 Sol rest on the stereochemistry argument and the shared protocol.

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5. Forward-verification elucidation

5.1 Method

The inverse direction is the model's hard, isomer-blind direction; the **forward** direction (structure→spectrum) is its easy, accurate one.[2] We close a training-free generator–verifier loop:

> **generate** candidate structures (inverse) → **forward-predict** each candidate's ^{13}C spectrum, blind to the observed spectrum → **re-rank** candidates by the distance between predicted and observed ^{13}C → return the best match.

Candidates are the inverse solver's proposals: 373 deduplicated structures across the 194 targets, forward-predicted in shuffled, anonymised batches, blind to the observed spectrum and the target's identity. Predicted and observed ^{13}C peak sets are then compared by symmetric chamfer distance (Fig. 6).

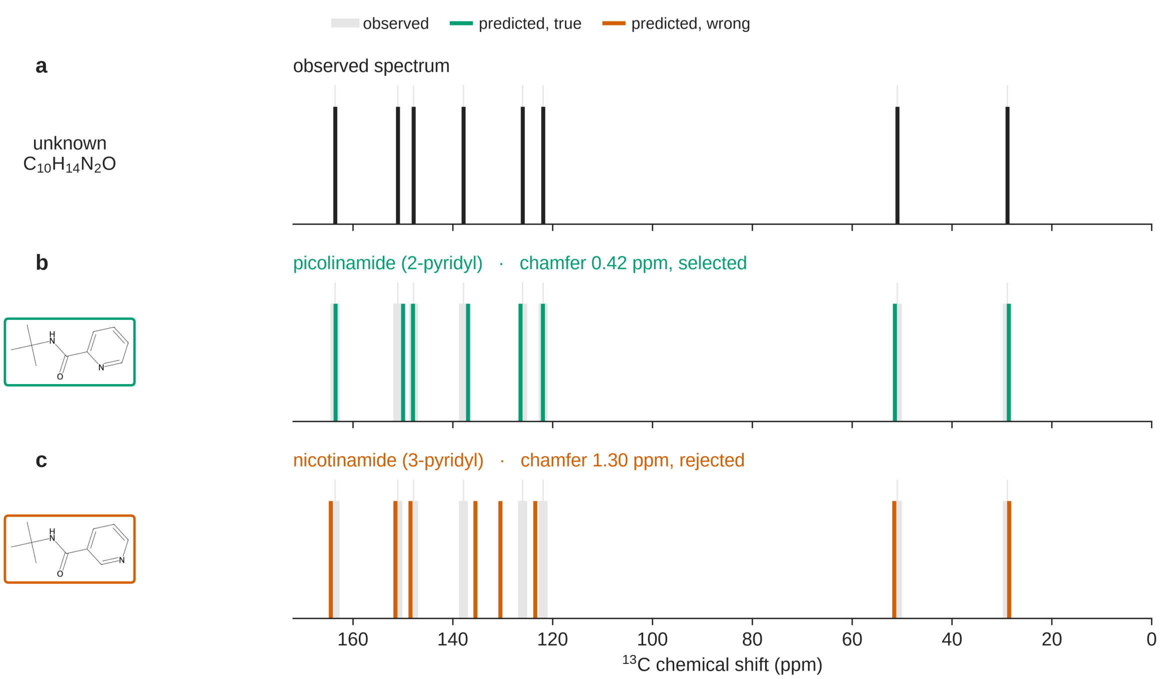

Fig. 6: Forward-verification on a benchmark regioisomer pair: picolinamide and nicotinamide are indistinguishable to the inverse task, but the true isomer's forward-predicted ^{13}C spectrum matches the observed one at a chamfer of 0.42 ppm against 1.30 ppm for the alternative (§5.1).

Regioisomers have *different* forward-predicted shifts, but the margin is thin: isomeric candidate pairs for the same target separate by a median **1.21 ppm** (quartiles 0.84 and 1.78), and **82% are predicted closer together than the predictor's own ~2 ppm error** — usually within the noise. The ranking nevertheless recovers the right one 89% of the time when it is present (§5.2), a real effect on a thin margin, confirmed against a derangement null at p=0.001 (§5.5).

Conditional precision is 72–89%, not 100%: near-degenerate candidates the predictor cannot resolve are its false positives. The same spacing sets limits §5 reports separately: the 54.0% derangement chance floor on multi-candidate compounds (§5.5) and the ceiling neither the HOSE lookup nor the GNN escapes (§5.4). Sharper predictors, not better search, would move them; they exist: ~1 ppm message-passing ensembles[36], a DFT-coupled graph network at 0.94 ppm[37], a community benchmark[38] and curated shift sets[39] for calibration.

The scheme is an analogy to DP4[15,16] and NMR crystallography[17,18], not an equivalence: we forgo DFT's ≈1–2 ppm accuracy and its calibrated, probabilistic error model for a predictor that runs in seconds.

5.2 Result

We ran the verifier over **every compound in the benchmark**. The first column below is the 60-compound arm (v3 + v2-control) on which §5.3–§5.5 build; the second is the full benchmark. All 373 candidates were forward-predicted, so nothing here is a lower bound; the self-ranking row re-derives Table 2 from §4's output.

Table 7. Forward-verification decomposition, original arm and full benchmark.

	60-compound arm	**full benchmark (n=194)**
generation recall	19/60 (32%)	**65/194 (34%)**
top-1, solver self-ranking	14/60 (23%)	55/194 (28%)
top-1, **forward-verified re-ranking**	16/60 (27%)	**58/194 (30%)**
conditional on recall — self-ranking	14/19 (74%)	55/65 (**85%**)
conditional on recall — **forward-verification**	16/19 (**84%**)	58/65 (**89%**)
…single-candidate compounds within that set	6/19	28/65
multi-candidate only — self-ranking	8/13 (62%)	27/37 (73%)
multi-candidate only — **forward-verification**	10/13 (77%)	**30/37 (81%)**

When the true structure is among the candidates, forward verification selects it in 58 of 65 cases (89%). Of the 65, **28 had a single candidate**, which any ranker scores by construction, so that pooled figure is not the verifier's margin over chance. On the **37 where a choice existed**, forward-verification gets **30/37 (81%)** against a 54.0% derangement floor (§5.5) — **+27.1 points**; self-ranking **27/37 (73%)**; we treat the 37 as the denominator that measures verification.

The margin over self-ranking remains small and unresolved: seven compounds gained, four lost, McNemar exact **p=0.55**. We do not claim it. The load-bearing claim is that precision is high **in absolute terms** while recall binds: top-1 moves only 28%→30% because the true structure was never proposed for 129 of 194 compounds, which no re-ranking can repair. The verifier converts recall almost completely on *simple* targets (50/53, **94%**) and is **exactly tied** with self-ranking on *complex* ones (8/12, 67%). Elucidation factorises into two near-independent levers: the **verifier is already strong (89%)**; the **generator (34% recall) is the wall** (Fig. 1).

5.3 Generate-wide: testing the recipe

The decomposition implies a recipe: *generate wide, verify by forward prediction* — a chemistry analog of self-consistency sampling.[40] Ten solver agents proposed up to **six regiochemistry-aware candidates per compound**, pooled with the originals and re-ranked as before.

Table 8. Generate-wide vs original, on the 60-compound arm only, since wide generation was run there; the "original" column is Table 7's first.

	original (60-compound arm)	generate-wide
recall (true structure among candidates)	32%	**42%**
forward-verified top-1	27%	**30%**
verification precision (conditional on recall)	84%	72%

Wide generation lifts recall +10 points (32%→42%, 19/60→25/60) and top-1 +7 over the self-ranking baseline (23%→30%, 14/60→18/60), i.e. +3 over the forward-verified top-1 (27%→30%, 16/60→18/60, Table 8), on the same 60 compounds (Fig. 7) with no training.

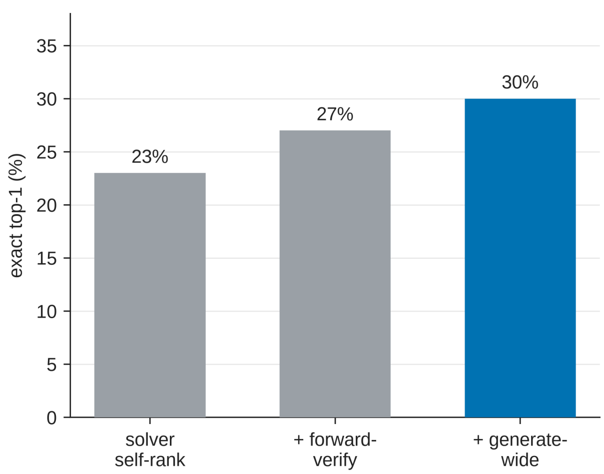

Fig. 7: The forward-verification inference ladder on the 60-compound arm: solver self-ranking → + forward-verification → + generate-wide, at 23% / 27% / 30% top-1. Each rung is a training-free change to inference alone, directional rather than individually resolved at this sample size (§5.3).

These top-1 differences are directional, not statistically resolved at n=60. The stages are **not** nested: self-ranking to generate-wide gains seven compounds and loses three, McNemar exact **p=0.34** (b=3, c=7). The recall gain is the better-supported effect: the ladder shows the *mechanism* works and recall is the movable factor, not how large the top-1 gain is.

The ladder does not extend. **(i) Recall plateaus at 42%**, on large polycyclic targets rather than exotic ones — **≥4 rings in 3.1% of recalled compounds against 38.0% of missed ones**. **(ii)**

Verification precision falls 84%→72% as more near-degenerate regioisomers enter the pool. This recall/precision tension is the honest ceiling of the training-free approach: closing the gap requires sharper verification or 2D-NMR constraints, not merely more candidates.

5.4 Non-LLM verifiers: a deterministic lookup and a learned model

§5.3 suggests an obvious fix: a non-LLM ^{13}C predictor in the verifier slot. We tested a HOSE-code[41]-style lookup and a small message-passing GNN, both trained on the same nmrshiftdb2 dump[42] and applied to the same §5.2 candidate sets, so that only the predictor changes. The GNN is deliberately modest; sharper purpose-built ^{13}C models exist.[36][37][38]

Table 9. Verifier comparison, conditional on recall.

verifier	60-compound arm (n=19)	full benchmark (n=65)	held-out ^{13}C MAE
solver self-ranking	14/19 (74%)	55/65 (85%)	—
deterministic HOSE *lookup*	14/19 (74%)	55/65 (85%)	3.23 ppm
learned GNN (same data)	16/19 (84%)	**59/65 (91%)**	**1.70 ppm**
LLM forward-verifier (§5.2)	16/19 (84%)	58/65 (89%)	—

The lookup ties the solver's own score at both sample sizes (Table 9). Coverage explains it: of the 6,360 candidate carbons, only **2%** match a training environment at r=4 and **71%** resolve at r≤2 or the hybridisation prior, too coarse to separate regioisomers.

The GNN matches and slightly exceeds the LLM verifier, 59/65 against 58/65 (Fig. S6), though not significantly: *suggestive and directional*, consistent with the lookup's failure being substantially **method** rather than coverage alone, but not established. Neither predictor reaches the near-degenerate-regioisomer precision ceiling (§5.3/§5.5), where DFT-level accuracy or orthogonal 2D-NMR constraints remain the genuine fix.

Leakage is slight — no benchmark answer appears in nmrshiftdb2 at all (0/373) — and a Y-randomisation control (1,000 derangements) places the real result above the 97.5th percentile of chance (n=19: real 84% vs mean 58.6%, one-sided $p<0.05$). Like §5.6, the learned verifier is a **trained complement**, reported outside the training-free protocol.

5.5 Negative control

Permutation negative control (Y-randomisation analog). A re-ranker exploiting genuine predicted-vs-observed ^{13}C agreement must lose it when the pairing is broken. We re-paired which observed ^{13}C spectrum each candidate set is scored against — a *derangement*, so no compound keeps its own spectrum — and re-ran the verifier 1,000 times. Conditional-on-recall precision falls from the true **89.2% (58/65)** to a permuted mean of **73.8%** (95% range 66.2–81.5%; one-sided empirical p=0.001, two-sided p=0.002). The verifier acts on real spectral agreement, not a candidate-list artefact. That floor is high because 28 of the 65 recall-positive compounds carry a single scorable candidate, and a compound with nothing to rank is scored correct under *every* pairing. Those compounds enter the permuted score as guaranteed hits while carrying no verification signal, so the control was partly measuring the composition of the recall-positive set. On the **37** multi-candidate compounds — the set §5.2 calls the one that measures verification — the floor falls to **54.0%** and the real value is **81.1% (30/37)**, a margin of **+27.1 points** rather than +15.4, at the same one-sided p=0.001. We report the restricted figure as the verifier's margin over chance and the pooled one for continuity with §5.2's denominator.

Confidence calibration (a negative result). Ranking the 138 multi-candidate compounds by chamfer margin and answering only the most-confident fraction leaves top-1 flat and non-monotonic with coverage (22% at full coverage, 24% at 75%, 28% at 50%, 24% at 25%). Single-candidate compounds, which have no margin, must be excluded; retaining them in an earlier analysis produced a spurious "improvement". We therefore do **not** claim the verifier distance as a calibrated abstention gauge (§5.4).

5.6 Is the recall wall task-intrinsic? A trained-generator probe

The ceiling above is a property of *training-free LLM elicitation*; whether the ~42% recall plateau is intrinsic to 1D-data elucidation is a separate question. We probed it by substituting a small (~16M-parameter) $^{1}H/^{13}C$→SMILES transformer for the enumerator; its candidates were pooled with Claude's and re-ranked by the same verifiers.

On the 194-compound benchmark the true structure enters the candidate pool for **54.1%** of compounds, versus 41.8% for scaffold enumeration and 33.5% for Claude alone. Where enumeration's near-degenerate regioisomers *collapse* the HOSE verifier (top-1 28.4%→16.0%, the §5.3 precision-loss mechanism), the generator's formula-correct, ^{13}C-separable candidates **convert**: top-1 rises **28.4%→35.1%** (McNemar exact p=0.015; +6.7 points, 95% CI [+2.1, +11.9]; Fig. S5). With the LLM forward-verifier on the 60 forward-verify compounds, recall rises 42%→**57%** (34/60) and top-1 reaches **47%** (28/60), at 82% precision conditional on recall (28/34). Those two figures are a **re-run, not a reproduction**: the earlier pass reported 41% top-1 at 73% precision but deposited no outputs, so we do not stand behind it, and the re-run lands above it.

Two controls guard against memorisation: the fine-tuning split removes every benchmark InChIKey-14 (train∩benchmark = 0, val∩benchmark = 0), with **none of the 40 newly-recovered compounds** in either training stage (0/40 in simulated pretraining, 0/40 in fine-tuning); and the simulated-pretrained model alone recovers **0/248** benchmark structures zero-shot, rising to 25% only after IRexp fine-tuning. The IRexp data, not the architecture, is the active ingredient, so the recall ceiling is **elicitation-specific, not task-intrinsic**. We report this as a probe, not part of the headline training-free protocol.

Purpose-trained systems make the point more strongly: NMR-Solver reaches 52.9% top-1 on experimental literature $^{1}H/^{13}C$ with the formula supplied[21], and a purpose-trained IR transformer 63.8% top-1 on experimental NIST spectra[14]. Whatever bounds a training-free LLM at 28–30%, it is plainly not the task. The public split `irexp_release/train` does *not* hold out the benchmark (117/200 InChIKey-14 overlap), so downstream users must de-leak; see Data and code availability.

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6. Discussion

The Claude-family models we tested are reliable **scaffold-level** elucidators and good **verifiers** (89% conditional on recall); exact top-1 is throttled by candidate recall and by the regiochemical underdetermination intrinsic to 1D NMR. The contribution is a **diagnosis with a bounded, training-free improvement attached**, not a method that solves the task.

The diagnosis held under every perturbation we were able to apply — four Claude models, a second chemical domain, four verifiers — and reproduces inside a single application domain (§4.5): the wall is **generation recall**, not verification, consistent with the bottleneck being structural rather than incidental. Whether it holds outside the Claude family is the open question, not a settled one (§7). We measured the improvement's ceiling rather than projecting past it: forward verification moves top-1 from 28% to 30% across the whole benchmark (§5.2), while recall plateaus at 42% (§5.3).

This reframes the engineering problem. The durable levers are not a bespoke spectra→structure model — a target the frontier already meets at the scaffold level, and one that ages out each model generation — but open, hard, honestly scored benchmarks and inference-time scaffolding that needs no training. Our trained-generator probe (§5.6) shows the recall wall is *elicitation-specific, not task-intrinsic*, and the gain comes entirely from the released data (0→25% with fine-tuning, 0/248 without): the **open dataset**, not a bespoke architecture, does the work when training helps.

Protocol is a lever of the same order as capability: a number quoted without its protocol is uninterpretable. We do *not* claim protocol dominates capability — the model axis spans 0% to 46% top-1 on the fixed 24-compound subset (Table 3), wider than the 5%→15% protocol effect in §4.3, and the two are measured on different sets. In molecular property prediction, a systematic benchmark reports that learned, pretrained molecular embeddings rarely outperform classical ECFP fingerprints once evaluation is controlled[43]; we read our forward-verification recipe (§5) in that light, and draw the parallel narrowly.

Two findings transfer to practice: solve each problem in bounded, frequently-reset contexts with tool access (5%→15%, 1/20→3/20, directional at n=20, McNemar $p \geq 0.25$, §4.3); and use forward-predicted-vs-observed ^{13}C agreement to *re-rank* candidates, not to decide whether to trust the winner. Match distance failed as an absolute confidence gauge in our selective-prediction test (§5.5, a reported null result), so it does not support abstention thresholds.

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7. Limitations

Several limitations of earlier drafts are now resolved; we state what remains plainly.

Resolved. **Extraction noise:** an RDKit self-consistency audit finds **57/60 ground truths spectrally clean**, with metrics unchanged on that subset. **Verifier sample size:** forward verification now runs over **all 194 compounds**, so the recall-conditional claim rests on n=65 rather than n=19 (§5.2). **Verifier precision:** precision degrades as near-degenerate regioisomers accumulate (§5.3), so forward-match distance is a strong *re-ranker* but a soft *confidence* gauge. **Non-LLM verifiers** (§5.4): a HOSE-code *lookup* does **not** beat the LLM verifier (85% vs 89% conditional at n=65) and a *learned* GNN reaches 91% — directional but not statistically resolved at n=65, so it is suggestive that the deterministic failure was method rather than coverage alone, not a demonstration of it. Beyond all of them lies the near-degenerate-regiochemistry precision ceiling, where DFT-level accuracy or 2D-NMR constraints are the genuine fix. **Projection:** §5.3 now measures what an earlier draft estimated: top-1 30%, recall 42%, below the estimate it replaces, reported as found rather than adjusted.

Independence checks. Scoring is mechanical RDKit, not LLM-judged, and all solver/verifier runs were transcript-audited *at generation time* for zero web and zero ground-truth access. That audit is the one part of the pipeline a reader **cannot re-verify from the release**: the committed artefacts are the parsed per-compound predictions, and the transcripts are not deposited (they are available on request). The *outcome* of closed-book solving is separately testable: the formula-only control (§4.6) removes the spectra and accuracy collapses, and a permutation negative control (§5.5) collapses conditional-on-recall precision from 89.2% to a chance mean of 73.8% (two-sided p=0.002 on the full benchmark), as a correctly isolated blind evaluation requires. A second Claude model (Sonnet) was comparably recall-bound on a 12-compound subset (recall 33% vs Opus 42% on those compounds), consistent with the recall bound being a property of the task rather than of one model — but within the Claude family, so it speaks to model-instance robustness, not cross-vendor generality.

Remaining. **(i) Pretraining contamination is bounded, not excluded.** Every benchmark compound was mined from open-access literature, so a frontier model may have encountered it in training. The formula-only control (§4.6) bounds how much of the headline number pure recall can explain: masking the spectra drops top-1 from 23% to 5%, perfectly nested, McNemar exact p=0.001. It examines the three formula-only successes individually rather than collectively — only one (an unlisted silyl aryl sulfonate) has a genuinely determining composition, while *N*-tosyl-leucine and [6]-paradol are catalogued compounds for which recall is a live explanation — which leaves the bound unchanged. A second control finds accuracy flat in source publication year across all 194 compounds (r=−0.007), the size-adjusted older-versus-newer difference bounded at −5.1 points, 95% CI [−17.2, +7.0]. Neither control is randomised: publication year is observational, and the formula-only arm cannot separate memorisation from inference on a near-determining formula. Nor is the recency test anchored to a *known* training cutoff, since the harness does not disclose one (§8); we test recency as a continuous proxy. A replication restricted to compounds published after a disclosed cutoff remains open.

The sharper form of this objection is one neither control addresses, and we state it plainly because a reader is entitled to weigh it. The prompt carries the spectral strings *exactly as printed in the source article*, down to its own typography — the sampler keeps a compound only when the raw ^{1}H payload can be recovered verbatim from the open-access full text (§8), so **every benchmark compound has its spectrum present verbatim in a document the model may have read**, and that string is a high-entropy fingerprint of the document. Masking the spectra therefore removes the chemistry and a retrieval key together: a collapse from 23% to 5% is what reading predicts, and also what retrieval predicts. Recency cannot separate them either, for the reason just given. The experiment that would is a spectrum meaning the same thing and reading differently — shifts perturbed within reporting precision and typography normalised, so the chemistry is untouched and no verbatim string survives. `scripts/jitter_control.py` builds that set from the released benchmark; running it is the obvious next control and we have not run it. It would bound verbatim retrieval only, not recognition from approximate shift patterns.

(ii) Human audit — prepared but not yet run. Two things are unvalidated by hand: the elucidation and forward-prediction outputs, and the *recall* side of dataset extraction (whether the parser found every IR string in every source paper). Transcription fidelity of the records we hold is measured and high (§2.3); record-level recall is not, and only reading papers can settle it. Solver and verifier are both LLMs, so the one validation we cannot perform ourselves is expert-chemist review. We have therefore **built and frozen** a blinded, pre-registered audit package at `data/audit/`, protocol at `docs/EXPERT_AUDIT_PROTOCOL.md`: a difficulty-stratified 30-compound elucidation panel plus seven ranking-only compounds, with a withheld answer key, testing what a miss actually is (§4) and whether forward verification is a trustworthy re-ranker (§5). **We have not yet run the panel**; until expert results are in, those claims should be read as machine-validated (RDKit InChIKey) but not yet human-validated. §4 has since narrowed its first question by reporting all 137 analysable misses mechanically (76.6% constitutional isomers, 22.6% scaffold-preserving positional errors), correcting the impressionistic claim the audit was built to check; what remains for a chemist is whether a formula-correct, scaffold-wrong candidate is a *chemically reasonable* reading of the spectra.

(iii) Single vendor and an underpowered cross-model comparison. The headline (28.4% top-1, n=194) is a single frontier model (Claude Opus) on the full benchmark; the cross-model evidence (§4.4) is four **Claude-family** models on a shared **24-compound** subset, and is underpowered: the ranking is robust (strictly nested, weakest floored at 0%) but no adjacent gap is established at n=24, so quantitative claims should be read as holding **for the Claude family**. The cross-**vendor** question is no longer open: §4.7 finds verification precision exceeding generation recall for Grok 4.6, Gemini 3.7 Flash and GPT-5.6 Sol on the 60-compound arm, as for Claude. What remains is weaker than the original gap but real: only Claude's inequality is interval-disjoint at n=60, the other three are directional. Those models were reached through a coding-assistant harness exposing reasoning-effort tiers, a decoding control our own arm never had, so they measure a model as served rather than a bare endpoint. The clean-clone contamination control was run for Grok alone, and the cross-vendor arms cover 60 compounds.

(iv) Domain-subset scope: the battery-electrolyte case study (§4.5) uses *literature compounds bearing electrolyte-relevant functional chemistry*, not operando or in-cell decomposition spectra, so it demonstrates functional-class transfer of the elucidation bottleneck, not direct assignment of authentic interphase/degradation products, which would require a dedicated operando set.

(v) Constitution-only scoring: correctness is judged on InChIKey connectivity, so a correct-constitution / wrong-stereochemistry prediction counts as correct. Scoring the full InChIKey gives 21.1% top-1 (§3), 7.3 points lower, so the headline should be read as an upper bound on full-stereochemistry accuracy. We keep constitution because 1D NMR/IR rarely determines absolute configuration — the 10.3% (20/194) of targets with a defined stereocentre would be penalised for information the prompt never carried — but a stereochemistry-sensitive benchmark would need 2D/chiroptical data.

(vi) Single-sample scoring: each headline compound is scored from one solver prediction set, so reported top-1/recall carry no run-to-run (LLM-sampling) variance estimate; the bootstrap CIs reflect compound sampling only. §5.3 pools ten independent generation passes (recall 32%→42%), indicating generator stochasticity that single-pass scoring understates.

(vii) Chemical-space coverage: the benchmark is drawn from open-access organic methodology and total-synthesis literature; it is **not** representative of organometallic/coordination compounds, large biomolecules (peptides, oligonucleotides, oligosaccharides), or stereochemistry-heavy targets, and our results should not be extrapolated to them.

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8. Methods

Mining and resolution. PMC-OA full text was fetched from `s3://pmc-oa-opendata`, parsed deterministically, and resolved with OPSIN, RDKit and SELFIES, with an optional cached PubChem fallback.

Benchmark and agents. Problems were sampled from `irexp_resolved`, stratified by RDKit ring analysis to half the round per stratum, and de-duplicated across rounds by InChIKey. A compound was admitted only if its raw ^{1}H payload — multiplicities and J values as printed — could be re-extracted verbatim from the PMC-OA full text of its source article (`benchmark_v2.raw_1h_for`), which is what puts genuine coupling information in the prompt and, unavoidably, puts the source article's exact string there too (§7). The battery-electrolyte subset (§4.5) was drawn from the same corpus by SMARTS filters for six electrolyte functional classes (carbonate, sulfonyl/sulfonate, nitrile, sp^{3}-C–F, phosphoryl, glyme/oligoether), balanced to eight compounds per class (48 curated; 46 scored after two yielded no parseable candidate), excluding every compound used elsewhere, and J-enriched and spectrally validated as in the main rounds. Solver and forward-prediction agents were independent Claude-Opus sub-agents under the same subscription, instructed closed-book and audited by automated transcript search for zero web/answer access. A prediction was scored correct on RDKit InChIKey-connectivity match; similarity used Morgan(2, 2048) Tanimoto, and forward-verification a symmetric chamfer distance over ^{13}C peak sets. The core protocol trains no model and uses no paid API; the §5.6 generator and the §5.4 learned ^{13}C verifier are the only two trained components, reported as complements and fenced from the headline results.

Models and versions. All experiments used Anthropic Claude models via the consumer subscription (claude.ai). The §4.4 comparison spanned, in capability order, Claude Haiku, Claude Sonnet, Claude Opus and Claude Fable 5; the headline benchmark (§4.1) and forward-verification (§5) used Claude Opus. `docs/MODELS.md` records, per experiment, the model, data directory, collection date, harness and tool access, and scoring code path, across three dated windows:

- **2026-06-09 to 2026-06-11** (UTC) — every candidate structure behind the **headline** results: §4.1, §4.3–§4.5, and the candidate pools all of §5 re-ranks.
- **2026-07-28** — the formula-only contamination control (§4.6), re-solving the same 60 compounds with spectra masked and generating its own candidates; it touches Table 5 alone.
- **2026-08-07** — three collections forward-predicting ^{13}C for candidates the June run had already produced (the §5.2 extension to all 194 compounds, the §5.3 coverage-gap closure, and the §5.6 re-run); none introduced a new candidate structure, and no recall number in the paper changes.

One *verified* number does: the §5.6 re-run supersedes a top-1 whose original forward-prediction outputs were never deposited and are lost, disagreeing with the figure it replaces (47% against 41%); §5.6 reports it as a re-run rather than a reproduction.

No model snapshot can be reported. The consumer harness exposes no checkpoint identifier, records nothing about which build served a given request, and exposes no decoding parameters, which were neither set nor recorded. `Claude Opus 4.8` is evidenced only for the 2026-06-09 pilot, so **a mid-window build change cannot be excluded** and we do not claim the same build served the main round two days later. Re-running reproduces the protocol distributionally rather than exactly, and exact agreement is not offered as a target. Fixed are the dated collection windows, the frozen per-compound outputs, and mechanical scorers that regenerate every number from them: exact reproducibility of scoring and analysis, not of inference (`docs/MODELS.md` §6).

Reproducibility. Every round is frozen: questions, ground-truth answers, per-agent raw outputs, predictions and scorer outputs are released; the sampler, scorer and forward-verification harness are scripted end-to-end.

Supporting Information figures

These are supplied as a separate Electronic Supplementary Information document (`docs/paper_esi.pdf`, built by `scripts/build_pdf.py`), as RSC requires; they are listed here for reference.

- **Fig. S1** (`docs/figures/fig0_overview.png`) — study design: open multimodal data (IRexp) → blind, complexity-stratified benchmark → decoupled blind solving → forward-verification re-ranking; training-free core pipeline.
- **Fig. S2** (`docs/figures/fig4_dataset.png`) — IRexp composition: IR records → NMR-paired → structure-linked → full IR+^{1}H+^{13}C+structure quadruples.
- **Fig. S3** (`docs/figures/fig2_size.png`) — accuracy vs molecular size (heavy-atom bucket); the monotonic 60%→7% top-1 gradient.
- **Fig. S4** (`docs/figures/fig6_electrolyte.png`) — top-1 and recovered accuracy on IRSpectra-Bench-Electrolyte by battery-electrolyte chemical class (n=46): sp^{3}-C–F easiest (50%), sulfonyl and nitrile hardest (12%); overall 26%/28%.
- **Fig. S5** (`docs/figures/fig_generator_probe.png`) — trained-generator probe (§5.6; a complement, not part of the training-free protocol): candidate recall and deterministic-HOSE top-1 on the 194-compound benchmark for Claude / + scaffold enumeration / + trained generator. Enumeration's near-degenerate isomers collapse the verifier (28.4→16.0%) while the generator's formula-correct candidates convert (28.4→35.1%).
- **Fig. S6** (`docs/figures/fig_verifier.png`) — learned-verifier probe (§5.4; a complement, not part of the training-free protocol). (A) Conditional-on-recall top-1 (n=65, whole benchmark) for the four verifiers: a GNN trained on the same nmrshiftdb2 data as the HOSE lookup reaches the LLM verifier's level (91% vs 89%) where the lookup (85%) does not move off the solver's own ranking. (B) Why: held-out ^{13}C MAE — the learned model is ~2× sharper (1.70 vs 3.23 ppm).

Data and code availability

Everything below is in the project repository. The archival deposit — a complete frozen snapshot of dataset, benchmark, answer keys, predictions, scripts, figure regeneration and the expert-audit package — will carry DOI [**TODO: 10.5281/zenodo.XXXXXXX — mint on submission**]; GitHub is the development mirror and the Zenodo record the citable version.

component	data	regenerated by
IRexp and its structure-complete split	`data/irexp/`, `data/irexp_resolved/`	`spectro_scraper/` (mining pipeline)
benchmark rounds and the within-compound control	`data/benchmark*/`	`scripts/benchmark_v2.py`
ground-truth integrity audit	`data/benchmark*/clean_qids.json`	`scripts/validate_benchmark.py`
headline accuracy (Table 2)	—	`scripts/score_main.py`

component	data	regenerated by
battery-electrolyte subset (§4.5)	`data/benchmark_electrolyte/`	`scripts/build_electrolyte_bench.py`, `scripts/score_electrolyte.py`
formula-only contamination control (§4.6)	`data/modality/`	`scripts/modality_ablation.py`
publication-recency control (§4.6)	`data/audit/recency_control.json`	`scripts/contamination_recency.py`
forward-verification, original arm (§5.2)	`data/fverify/`	`scripts/forward_verify.py`
…extended to all 194 compounds	`data/fverify_main/`	`scripts/forward_verify_main.py`, `scripts/forward_verify_all.py`
generate-wide arm (§5.3)	`data/gw/`, `data/fverify2/`	`scripts/score_generate_wide.py`, `scripts/ladder_significance.py`
…its coverage gap, closed	`data/fverify_gw/`	`scripts/forward_verify_gw.py`
non-LLM verifier comparison (Table 9)	`data/fverify/hose_results.txt`, `data/fverify/verifier_table_results.txt`	`scripts/hose_predict.py` (incl. `coverage`), `scripts/verifier_table.py`, `scripts/verifier_leakage.py`
negative control and selective prediction (§5.5)	—	`scripts/verifier_diagnostics.py`
what a miss is, and isomer separability (§4.1, §5.1)	—	`scripts/analyze_misses.py`, `scripts/isomer_separability.py`
difficulty-threshold sensitivity (§3)	—	`scripts/difficulty_sensitivity.py`
licence-pool split (§2.1)	—	`scripts/split_license_pools.py`
recall headroom and scaffold enumeration	—	`scripts/analyze_recall_headroom.py`, `scripts/enumerate_isomers.py`, `scripts/closing_the_gap.py`
blinded expert-audit package (§7)	`data/audit/`	`scripts/make_audit_sample.py`
manuscript integrity gates	—	`scripts/check_manuscript.py`, `scripts/verify_statistics.py`

Trained complements, both fenced from the training-free protocol. The §5.6 generator ships as a self-contained bundle at `contrib/generator_probe/` — candidates, the de-leaked split with its InChIKey-14 manifest, the verification scripts (`scripts/closing_the_gap_gen.py`, `scripts/forward_verify_gen.py`, `scripts/verify_leakage_exact40.py`) and the blind forward-prediction outputs behind its verified top-1 (`data/fverify_gen/`), so its scorer runs with no missing predictions. The §5.4 learned verifier ships `scripts/gnn_predict.py` (extract / train / score / control), the trained model `data/nmrshiftdb/gnn_c13.pt`, per-compound results and both leakage checks (`data/fverify/gnn_results.txt`), and the write-up `docs/VERIFIER_PROBE.md`. Model checkpoints are deposited on Zenodo. Both trained predictors of §5.4 need the nmrshiftdb2 dump, which we cannot redistribute; the README gives the one-line fetch.

Companion technical notes: `docs/BENCHMARK.md`, `docs/FORWARD_VERIFY.md`, `docs/MODALITY_ABLATION.md`, `docs/EXPERT_AUDIT_PROTOCOL.md`, `docs/MODELS.md`.

Two cautions for re-users. The public `irexp_release/train` split does **not** hold out IRSpectra-Bench — it overlaps by 117/200 InChIKey-14 — so de-leak with `contrib/generator_probe/build_exp_manifest.py` before training anything that will be evaluated on the benchmark. And the held-out answer keys `data/audit/key.jsonl` and `data/modality/key.json` are deposited but flagged *withhold from blinded reviewers*.

Licensing and attribution

IRexp is derived from two open-access source pools and redistributed under terms compatible with each (see `data/NOTICE`). **(a) Redistribution:** we release only *extracted numeric data* — IR band lists, $^{1}H/^{13}C$ shift lists, and resolved structures (SMILES/SELFIES/InChIKey) — plus each record's source DOI/accession, not source full text, figures, or PDFs. **(b) Two separable pools:** 119,345 records derive from the PMC Open-Access Subset (**CC-BY-4.0**) and 1,888 from the Chemotion RADAR4Chem FT-IR deposit (**CC-BY-SA-4.0**). The two are separable losslessly by `source_doi` — Chemotion records carry the `10.22000` prefix — and `scripts/split_license_pools.py` materialises them as two files with each record stamped `license`, so users may take the CC-BY pool alone; any combined or Chemotion-derived release carries CC-BY-SA-4.0 to honour the ShareAlike term. Code is released under the MIT License. **(c) Attribution:** re-users must cite this dataset (Zenodo DOI above) and attribute the original publications via each record's `source_doi`.

Author Contributions

I.Y.: conceptualization, methodology, software, formal analysis, investigation, data curation, visualization, writing — original draft. **R.S.:** methodology, software, formal analysis, investigation, validation (trained-generator and learned-verifier probes, §5.4 and §5.6), writing — review and editing. **R.A.V.-H.:** conceptualization, methodology, supervision, writing — review and editing.

Conflicts of Interest

The authors declare no competing interests.

Use of AI tools

This work studies a large language model, and LLMs were also used as instruments and as writing aids; we state both roles explicitly. **As an object of study:** all reported elucidation, forward-prediction and verification results were produced by Claude models invoked under the protocol of §3 and §8 — these are the measurements the paper reports, not assistance in producing it. **As a writing aid:** the authors used an LLM-based coding assistant for figure generation, analysis scripting, manuscript copy-editing, and for the internal review pass that produced several of the corrections recorded in the repository history. **No text, number, figure or citation in this**

manuscript was accepted without author verification against the released data and code; every quantitative claim regenerates from the scripts in `scripts/`. The authors take full responsibility for the content.

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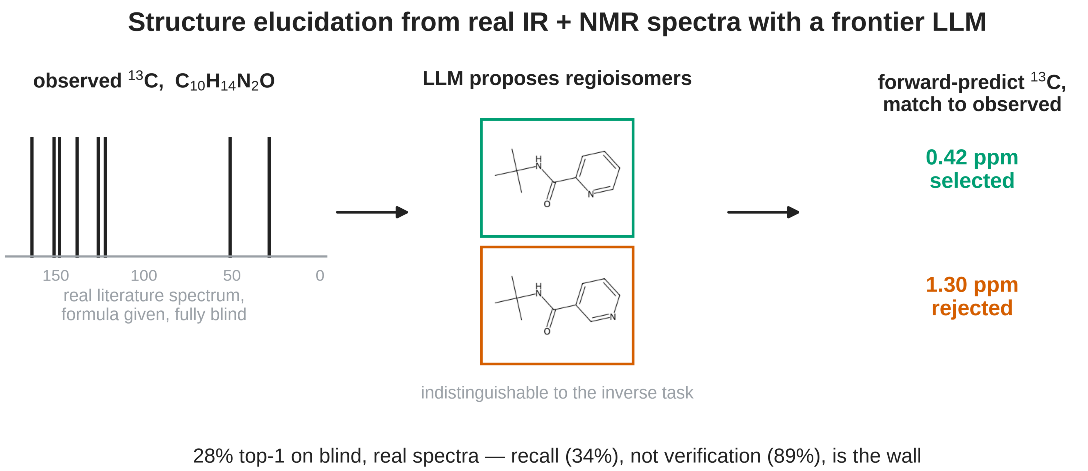

Recall, not verification, is the wall. On real, blind IR + ^{1}H + ^{13}C literature spectra a frontier LLM proposes the true structure for 34% of 194 compounds and, once proposed, selects it 89% of the time — lifting top-1 from 28% to 30%.